

Sebastian Herrera Pulido
AI in Construction and Facilities Management
Dr. Jake Smithwick
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SUMMARY:

1. Main Findings and Arguments

a. Limitations of traditional durability design

- Conventional prescriptive design methods (fixed w/c ratios, cement content, cover depth) are widely used but often fail to capture real degradation mechanisms such as carbonation, chloride ingress, freeze–thaw cycles, and chemical attack.
- These methods are particularly problematic for low-carbon concretes (e.g., GGBFS, fly ash, calcined clays), whose durability behavior differs significantly from ordinary Portland cement but is not adequately addressed by existing standards.

b. Role of computational modelling

- Computational durability models (transport, reactive transport, and damage models) provide more realistic, physics-based predictions of long-term concrete performance.
- Models based on Fick’s laws, finite-element methods, and multi-physics frameworks can simulate carbonation depth, chloride profiles, sulphate attack, freeze–thaw damage, and coupled degradation processes over a structure’s service life.
- These approaches enable performance-based design (PBD), shifting focus from material prescriptions to predicted in-service performance.

c. Integration of Artificial Intelligence (AI)

- AI and machine/deep learning models improve prediction accuracy by handling large datasets and capturing non-linear relationships between materials, environment, and degradation.
- AI helps bridge the gap between laboratory tests and real in-service behavior, though most current studies still focus on isolated mechanisms rather than fully integrated durability systems.

d. Coupling durability modelling with Life-Cycle Assessment (LCA)

- A key contribution of the paper is advocating durability-informed LCA.
- Traditional LCAs rely on assumed service lives and repair intervals; by contrast, integrating durability models allows service life, maintenance frequency, and embodied carbon to be calculated based on actual degradation mechanisms.

- This coupling provides a more accurate estimate of true environmental impacts over a structure's lifetime.

2. Broader Implications for Society

- More sustainable infrastructure: Accurate durability predictions reduce premature failures, unnecessary repairs, and over-design, lowering long-term resource use and carbon emissions.
- Improved climate accountability: By linking durability to LCA, carbon accounting becomes more realistic, supporting climate targets and informed policy decisions.
- Resilience to climate change: Advanced models can account for changing environmental conditions, helping infrastructure adapt to harsher or more variable exposure regimes.
- Alignment with global goals: The framework supports UN Sustainable Development Goals, particularly SDG 9 (Industry, Innovation, Infrastructure) and SDG 12 (Responsible Consumption and Production).

3. Relevance and Potential Influence on the Construction Industry

a. Shift toward performance-based design

- Encourages designers, owners, and regulators to move away from “deemed-to-satisfy” rules toward quantified service-life predictions.
- Supports more rational decisions on cover depth, material selection, and maintenance planning.

b. Safer adoption of low-carbon concretes

- Provides tools to validate durability of low-carbon binders, reducing industry reluctance and enabling broader adoption without increasing long-term risk.

c. Cost and risk reduction

- Probabilistic and AI-enhanced models help predict deterioration timelines, allowing proactive maintenance, reduced life-cycle costs, and improved asset management.

d. Digital transformation of construction

- Integration with AI, BIM, and computational platforms positions durability modelling as part of digital design and decision-support systems.
- Facilitates collaboration between designers, contractors, and material suppliers through performance-based specifications.

Overall takeaway

The article argues that future concrete infrastructure must be designed using integrated computational durability models, AI, and life-cycle assessment. This approach enables structures that are not only low-carbon at construction but also durable, resilient, and environmentally efficient over their entire service life, representing a significant paradigm shift for the construction industry.